

ROCK HILL, SC, USA

WMO: 723117

Lat: 34.987N

Lon: 81.058W

Elev: 669

StdP: 14.34

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 53871

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.3	24.1	2.6	6.4	29.5	7.3	8.2	30.7	19.7	52.0	17.8	51.2	3.3	340	0.384

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	94.7	74.5	92.5	74.3	90.5	74.0	77.9	88.5	77.0	87.4	76.1	85.9	6.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.1	135.2	81.2	74.1	130.2	80.3	73.2	126.4	79.6	41.6	88.6	40.7	87.6	39.8	86.0	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.9	14.8	12.3	DB	13.5	98.0	3.9	3.0	10.7	100.1	8.5	101.9	6.3	103.5	3.5	105.7	
			WB	12.0	79.3	3.9	1.1	9.2	80.2	6.9	80.8	4.7	81.5	1.8	82.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.6	41.9	45.2	52.6	61.5	69.3	77.2	79.7	78.7	73.3	62.0	51.4	45.3
	DBStd	15.28	9.95	8.92	9.13	7.50	6.35	4.25	3.40	4.01	5.35	7.73	8.14	8.68
	HDD50	852	292	187	80	7	0	0	0	0	0	6	79	201
	HDD65	3037	718	557	394	152	36	0	0	0	7	149	413	611
	CDD50	5082	40	52	161	351	600	815	921	889	699	377	120	57
	CDD65	1792	0	2	10	47	171	366	456	424	257	54	4	1
	CDH74	16221	2	11	110	532	1482	3444	4357	3830	1984	435	31	3
	CDH80	6707	0	0	12	116	474	1515	2036	1705	753	94	2	0
Wind	WSAvg	5.0	5.6	5.8	6.1	6.2	5.3	4.6	4.5	3.8	4.2	4.3	4.6	4.8
Precipitation	PrecAvg	46.9	4.3	4.0	4.8	3.2	3.4	4.3	4.1	4.2	4.2	3.5	3.3	3.5
	PrecMax	67.7	8.1	9.0	10.2	8.0	8.9	10.4	9.9	13.8	12.2	11.8	8.3	9.9
	PrecMin	29.3	0.4	0.9	0.7	0.7	0.5	0.5	0.4	0.5	0.0	0.0	0.5	0.5
	PrecStd	7.9	2.0	2.0	2.2	1.9	1.9	2.3	2.3	2.7	3.0	2.7	1.7	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.1	75.8	81.8	86.8	91.4	96.6	97.3	97.6	93.6	87.3	78.5	72.8
		MCWB	62.1	62.8	63.6	66.0	72.2	73.9	75.7	74.1	71.3	72.7	64.7	63.7
	2%	DB	66.3	69.7	77.2	83.4	88.2	93.4	94.6	93.8	90.7	82.1	73.2	68.1
		MCWB	58.8	57.7	61.0	65.7	70.5	73.9	75.4	74.6	71.4	68.4	61.5	61.6
	5%	DB	62.8	65.4	73.0	80.4	85.2	90.9	92.2	91.2	87.9	79.1	70.0	64.4
		MCWB	56.2	55.6	59.0	63.6	69.3	73.7	75.2	74.9	71.6	66.1	60.8	59.2
	10%	DB	58.3	61.3	68.6	76.6	82.2	88.3	90.0	88.8	84.4	75.1	66.1	60.9
		MCWB	52.1	52.6	56.5	62.2	68.4	72.8	74.8	74.3	71.0	65.5	58.3	55.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.4	65.5	65.9	70.8	74.9	77.7	79.3	79.0	77.3	75.7	69.2	67.3
		MCDB	68.8	72.3	75.7	78.7	84.8	89.6	90.1	88.7	85.5	80.6	71.7	69.7
	2%	WB	61.3	62.1	63.8	68.7	72.9	76.5	78.0	77.8	75.5	72.8	65.8	63.7
		MCDB	65.0	67.5	72.4	77.5	82.4	88.1	89.3	87.7	83.3	77.9	69.6	66.3
	5%	WB	57.8	58.3	61.7	66.8	71.5	75.3	77.1	76.8	74.2	70.1	63.3	60.5
		MCDB	61.7	63.5	69.8	75.6	81.2	86.5	88.2	86.5	81.4	75.1	67.2	63.7
	10%	WB	53.3	54.1	59.5	64.5	70.2	74.4	76.1	75.8	72.9	67.6	60.2	56.2
		MCDB	57.2	59.1	65.9	72.8	79.1	84.8	86.4	85.0	79.6	73.6	65.2	59.4
Mean Daily Temperature Range	5% DB	MDBR	20.4	21.1	22.6	23.7	21.3	20.4	19.6	18.9	19.2	21.8	22.8	20.1
		MCDBR	24.3	25.9	26.8	27.5	24.4	23.7	22.5	22.3	24.3	24.8	26.1	22.5
		MCWBR	16.6	15.9	13.6	12.1	9.5	7.9	7.1	7.1	8.4	11.7	15.4	15.8
	5% WB	MCDBR	20.3	21.3	22.3	22.2	20.0	20.6	19.5	18.9	18.1	18.5	19.0	18.2
		MCWBR	16.2	15.6	13.2	11.1	8.5	7.7	7.1	6.9	7.5	10.3	14.1	14.6
Clear-Sky Solar Irradiance	taub	0.315	0.324	0.361	0.405	0.461	0.487	0.538	0.518	0.432	0.385	0.334	0.315	
	taud	2.498	2.468	2.394	2.264	2.149	2.116	1.997	2.039	2.265	2.373	2.476	2.530	
	Ebn at Noon	283	292	289	279	264	256	242	245	262	266	272	276	
	Edh at Noon	26	30	36	43	49	50	56	53	40	33	27	24	
All-Sky Solar Radiation	RadAvg	845	1049	1388	1735	1922	2035	1906	1745	1488	1228	932	728	
	RadStd	48	119	101	134	157	146	125	83	131	150	76	68	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (67 neighbors)	+0.95	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+306	+211

Nomenclature: See separate page